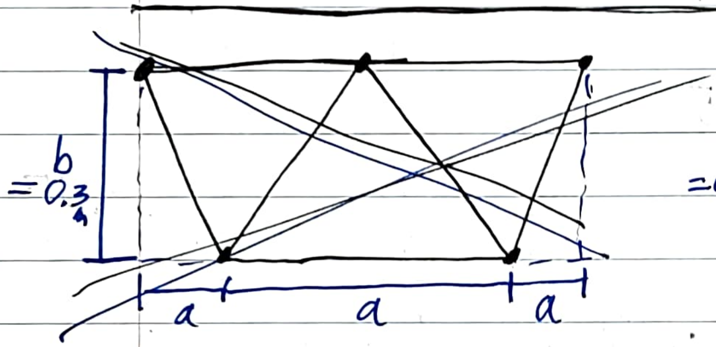
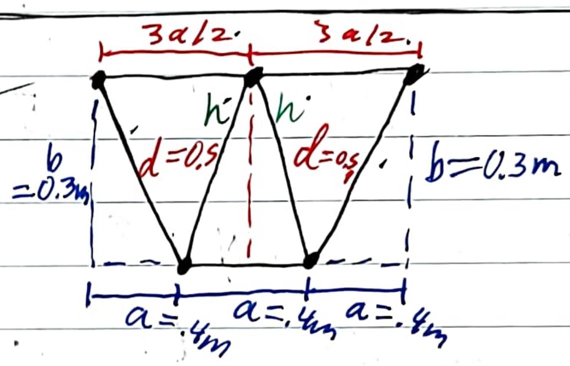
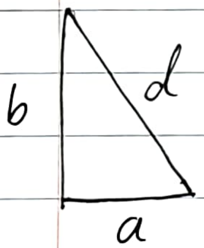
Density (ρ) of Grade A A500 steel:

$\rho = 7{,}800\ kg/m^3$







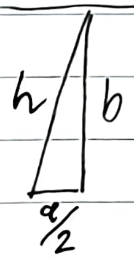
$d = \sqrt{a^2 + b^2}$

$d = \sqrt{(0.4)^2 + (0.3)^2}$

$d = 0.5m$

$a = 0.4m$

$b = 0.3m$



$h = \sqrt{\left(\frac{a}{2}\right)^2 + b^2}$

$h = \sqrt{\left(\frac{0.4}{2}\right)^2 + (0.3)^2}$

$h = 0.360555127 5m$

$\frac{3a}{2} = \frac{3(0.4)}{2} = 0.6m$

2 ~~Bars~~ Beams Length d (0.5m)
2 Bars Length $^{3a}/_{2}$ (0.6m)
2 Beams Length h (0.3606m)
1 Beam Length a (0.4m)

Diameter = 20 mm | radius = 10 mm

2 Beams of Length $d = 0.5m$

$0.5m \cdot \frac{1000mm}{1m} = 500mm$

volume of a cylinder $V = \pi r^2 h$

$V = \pi (10mm)^2 (500mm) = \boxed{157{,}079.63 mm^3}$

$\rho = 7800 \, Kg/m^3$

$157{,}079.63mm^3 \cdot \frac{1m}{(1000mm)^3}$

$weight = \rho \cdot V = (7800 \frac{Kg}{m^3})(157{,}079.63 \cdot 10^{-9} m^3)$

$weight = \boxed{1.225 \, Kg}$ per beam

weight of Both: $2 \cdot 1.225 Kg = \boxed{2.450 Kg}$

2 Beams of Length $\frac{3a}{2} = 0.6m$

$0.6m \cdot \frac{1000mm}{1m} = 600mm$

$V = \pi r^2 h$

$V = \pi (10mm)^2 (600) = \boxed{188{,}495.56 \, mm^3}$

$weight = \rho \cdot V = 7800 \frac{Kg}{\cancel{m^3}} \cdot 188{,}495.56 \cdot 10^{-9} \cancel{m^3}$

$weight = \boxed{1.470 \, Kg}$ per beam

wieght of both beams: $1.470 \cdot 2 = \boxed{2.940 Kg}$

2 Beams of Length $h = 0.3606\,m$

$$0.3606\,m \cdot \frac{1000\,mm}{1\,m} = 360.6\,mm$$

$$V = \pi r^2 h = \pi (10\,mm)^2 (360.6\,mm) = \boxed{113{,}285.83\ mm^3}$$

$$\text{weight} = \rho \cdot V = 7800\,\frac{kg}{m^3} \cdot 113{,}285.83 \cdot 10^{-9}\,m^3$$

$$\text{weight} = \boxed{0.884\ kg} \text{ per beam}$$

$$\text{weight of both beams: } 0.884 \cdot 2 = \boxed{1.768\ kg}$$

---

1 Beam of Length $a = 0.4\,m$

$$0.4\,m \cdot \frac{1000\,mm}{1\,m} = 400\,mm$$

$$V = \pi r^2 h = \pi (10\,mm)^2 (400\,mm) = \boxed{125{,}663.7\cancel{06}\ mm^3}$$

$$\text{weight} = \rho \cdot V = 7800\,\frac{kg}{m^3} \cdot 125{,}663.71 \cdot 10^{-9}\,m^3$$

$$\text{weight} = \boxed{0.980\ kg} \text{ per beam}$$

– Only 1 beam

Max weight of just the beams:

$$2.45 + 2.94 + 1.768 + 0.980 = \boxed{8.138\ kg}$$